

BAND-LIMITED PULLBACKS OF HARDY Z , ORTHOGONAL SPECTRAL MEASURES, EDGEWORTH CUMULANTS, AND SPECTRAL LAGUERRE POSITIVITY

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ABSTRACT. The normalized pullback of Hardy's Z -function by the Riemann–Siegel phase is a finite-power stationary vector with sine-kernel covariance $K(h) = \sin h/h$. Band-limitedness and the orthogonal spectral representation $X(u) = \int_{-1}^1 e^{iu\xi} \Phi(d\xi)$ follow. The stationary pullback extends to a real entire function of order one and type one, and its generalized Laguerre functionals admit the spectral form

$$L_n[X](u) = \frac{1}{2} \int_{[-1,1]^2} (\xi - \eta)^{2n} e^{iu(\xi+\eta)} \Phi(d\xi) \Phi(d\eta).$$

Membership in the Laguerre–Pólya class $\mathcal{LP}$ is equivalent to positivity of all $L_n[X](u)$, $n \geq 1$, $u \in \mathbb{R}$. The zero correspondence between X and ζ on the critical line is exact, so RH is unconditionally equivalent to this spectral Laguerre positivity. The Gaussian realization is quasi-free; the Laguerre kernels are symplectic generators with metaplectic character $\det(I - 2itA_{n,u})^{-1/2}$, giving the cumulant formula $\frac{(m-1)!}{2} \text{Tr}(A_{n,u}^m)$.

1. THE NORMALIZED PULLBACK AND THE COVARIANCE THEOREM

Definition 1.1 (Phase pullback). Let

$$\theta(t) = \arg \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi$$

be the Riemann–Siegel phase. On the real line where $\Theta = \theta$ is strictly increasing for t large, put

$$u = \Theta(t), \quad t = t(u) = \Theta^{-1}(u).$$

The variance-normalized pulled-back Hardy signal is

$$X(u) = \frac{Z(t(u))}{\sqrt{2\Theta'(t(u))}}, \quad Z(t) = e^{i\theta(t)} \zeta\left(\frac{1}{2} + it\right).$$

Theorem 1.2 (Sine-kernel covariance).

$$K(h) := \lim_{U \rightarrow \infty} \frac{1}{U} \int_0^U X(u+h)X(u) du = \frac{\sin h}{h}, \quad K(0) = 1.$$

Proof. Step 1: Scaling. On $t \in [T, 2T]$, $\theta'(t) = \frac{1}{2} \log(t/2\pi) + O(t^{-1})$; set $L := \frac{1}{2} \log(T/2\pi)$ and $N := \lfloor \sqrt{T/2\pi} \rfloor$, so $U = \Theta(2T) - \Theta(T) \asymp TL$. Under $u = \Theta(t)$ with $t_h := \Theta^{-1}(\Theta(t) + h)$, Taylor expansion gives $t_h = t + h/\Theta'(t) + O(h^2/(TL^2))$ and the prefactor is $\frac{1}{2}(1 + O(1/(tL)))$. Hence

$$K(h) = \lim_{T \rightarrow \infty} \frac{1}{2TL} \int_T^{2T} Z(t_h)Z(t) dt + O(1/L).$$

Step 2: Four signs. Write $Z = S + \bar{S}$ with $S(t) := \sum_{n \leq N} n^{-1/2} e^{i\theta(t) - it \log n} + R_1(t)$ and $\|R_1\|_{L^2[T, 2T]}^2 = O(T^{1/2} \log T)$ (Titchmarsh, Ch. VII). The product $Z(t_h)Z(t)$ splits into four sign-pieces plus R_1 -cross terms; the latter contribute $O(T^{3/4}(\log T)^{1/2})$, i.e. $O(T^{-1/4}/L)$ after division by TL .

Step 3: Mixed-sign via Montgomery–Vaughan. For $S_h \bar{S}$ the phase becomes $ih(1 - \log n/L) + it \log(m/n) + O(1/L)$. MV gives off-diagonal $O(N) = O(T^{1/2})$, and the diagonal yields $TL \int_0^1 e^{ih(1-x)} dx + O(T)$; the conjugate mixed piece gives the conjugate. The Riemann-sum rate is $O(1/L)$ uniformly for h in compact sets by Abel summation against $\sum_{n \leq y} 1/n = \log y + \gamma + O(1/y)$.

Step 4: Same-sign via Atkinson. For $S_h S$ the phase is $2i\theta(t) + ih + O(1/L)$; the functional equation $\chi(\frac{1}{2} + it) = e^{-2i\theta(t)}$ converts $e^{2i\theta(t)} \zeta^2$ into a χ^{-1} -twisted second moment, which by Atkinson's formula (Ivić, Ch. 15) is $O(T^{1/2} \log T)$, i.e. $O(T^{-1/2} \log T/L)$ after division by TL .

Step 5. $\frac{1}{2TL} \int_T^{2T} Z(t_h) Z(t) dt = \int_0^1 \cos(h(1-x)) dx + O(T^{-1/4} + L^{-1}) \rightarrow \sin h/h$. $\square$

2. THE CYCLIC HILBERT SPACE AND TRANSLATIONS

Definition 2.1. $\mathcal{H}_X = \overline{\text{span}\{X(\cdot + h) : h \in \mathbb{R}\}}$ with $\langle f, g \rangle_W = \lim_U \frac{1}{U} \int_0^U f \bar{g} du$; cyclic vector $x_0 = X$.

Proposition 2.2 (Translations are unitary). $(U_a f)(u) = f(u + a)$ is a strongly continuous unitary group on $\mathcal{H}_X$, and $\langle U_h x_0, x_0 \rangle_W = K(h)$.

Proof. Wiener-mean translation invariance gives the isometry; U_{-a} is its inverse. Strong continuity is continuity of K plus cyclic density. $\square$

3. BAND-LIMITEDNESS

Theorem 3.1 (Spectral representation). *There exists a projection-valued measure E with $U_h = \int e^{ih\xi} E(d\xi)$ and $\Phi(B) := E(B)x_0$ satisfies $\langle \Phi(A), \Phi(B) \rangle_W = \mu(A \cap B)$ with $\mu(B) = \|E(B)x_0\|_W^2$; moreover $X(u) = \int e^{iu\xi} \Phi(d\xi)$.*

Theorem 3.2 (Band-limitedness). $\mu(d\xi) = \frac{1}{2} \mathbf{1}_{[-1,1]}(\xi) d\xi$, and $X(u) = \int_{-1}^1 e^{iu\xi} \Phi(d\xi)$.

Proof. $\sin h/h = \int_{-1}^1 e^{ih\xi} d\xi/2$; Bochner uniqueness identifies μ . For $B \cap [-1, 1] = \emptyset$, $\|\Phi(B)\|_W^2 = \mu(B) = 0$. $\square$

4. ENTIRE CONTINUATION, ORDER AND TYPE

This section builds the entire function on $\mathbb{C}$ that extends the real-line stationary process X , records its order and type, and matches it to the hypothesis class of the Csordas–Dimitrov generalized Laguerre criterion.

Proposition 4.1 (Paley–Wiener extension). *The $\mathcal{H}_X$ -valued map*

$$X(z) := \int_{-1}^1 e^{iz\xi} \Phi(d\xi), \quad z \in \mathbb{C},$$

is entire in z , and for every $z \in \mathbb{C}$,

$$\|X(z)\|_W \leq e^{|\text{Im } z|} \sqrt{\|\Phi([-1, 1])\|_W^2} = e^{|\text{Im } z|}.$$

In particular, the scalar function X on the real line extends to an entire function of exponential type at most one, of order one and type exactly one.

Proof. Entirety is differentiation under the spectral integral, legitimate by $\|\Phi\|_W < \infty$ on the compact support $[-1, 1]$. The bound is

$$\|X(z)\|_W^2 = \int_{-1}^1 |e^{iz\xi}|^2 \mu(d\xi) = \int_{-1}^1 e^{-2\xi \text{Im } z} \frac{d\xi}{2} \leq e^{2|\text{Im } z|}.$$

Sharpness of type one: take $z = iy$; then $\|X(iy)\|_W^2 = \frac{\sinh 2y}{2y} \sim \frac{e^{2y}}{4y}$ as $y \rightarrow +\infty$. $\square$

Proposition 4.2 (Realness on the real line). *The scalar function X is real on $\mathbb{R}$ and therefore its entire continuation satisfies $X(\bar{z}) = \overline{X(z)}$.*

Proof. $X(u)$ is $\mathbb{R}$ -valued because $Z(t)$ is real for $t \in \mathbb{R}$ (Hardy's Z) and $\Theta' > 0$ makes the normalization real. Entire functions agreeing with a real function on $\mathbb{R}$ satisfy the Schwarz reflection identity. $\square$

Corollary 4.3 (X is in the Hermite–Biehler setting). *X is a real entire function of exponential type one in the finite-order Laguerre–Pólya ambient class. In particular the Csordas–Dimitrov generalized Laguerre criterion applies to X .*

Proof. Order one and finite type place X in Levin's class of real entire functions of exponential type, which is contained in the ambient finite-genus class of Csordas–Dimitrov. $\square$

5. THE CRAMÉR INVERSE

Theorem 5.1 (Inverse for increments). *For $B = (a, b]$, $\Phi(B) = L_W^2\text{-}\lim_R \frac{1}{2\pi} \int_{-R}^R \frac{e^{-iau} - e^{-ibu}}{iu} X(u) du$.*

Proof. Truncated Fourier inverses $P_R \mathbf{1}_B \rightarrow \mathbf{1}_B$ in $L^2(\mu)$; isometry gives the conclusion. $\square$

Proposition 5.2 (No point-frequency inverse). *For $\lambda \in (-1, 1)$, $I_T(\lambda) = \pi^{-1} \int_0^T e^{-iu\lambda} X du$ satisfies $\|I_T\|_W^2 \sim T/\pi$ and is not Cauchy.*

6. GAUSSIAN REALIZATION AND QUASI-FREENESS

Definition 6.1. $W : \mathcal{H}_X \rightarrow L^2(\Omega)$ isonormal: $\mathbb{E}[W(f)W(g)] = \langle f, g \rangle_W$, and $\Phi_G(B) := W(\Phi(B))$.

Theorem 6.2 (Quasi-freeness). *$\text{cum}(W(f_1), \dots, W(f_r)) = 0$ for $r \geq 3$.*

Proof. Gaussian CGF is quadratic; derivatives of order ≥ 3 vanish. $\square$

7. HEISENBERG/METAPLECTIC STRUCTURE ON THE BAND

7.1. Translation generator. $U_a = e^{iaP}$ with P unitarily equivalent to multiplication by ξ on $L^2([-1, 1], \frac{1}{2}d\xi)$ under the isometry $\Phi: f \mapsto \int f d\Phi$. The algebra $\text{span}\{P\}$ is abelian: $e^{iaP}e^{ibP} = e^{i(a+b)P}$.

7.2. Heisenberg–Weyl algebra.

Definition 7.1. On $\mathcal{D} := C_c^\infty((-1, 1)) \subset L^2([-1, 1], \frac{1}{2}d\xi)$, $Q := i\partial_\xi$.

Proposition 7.2 (CCR). *$[Q, P] = i\mathbf{1}$ on $\mathcal{D}$; the real span $\mathfrak{h}_3 = \text{span}_{\mathbb{R}}\{Q, P, \mathbf{1}\}$ is the Heisenberg algebra.*

Theorem 7.3 (Weyl relation). *$e^{i(aP+bQ)} = e^{-iab/2}e^{iaP}e^{ibQ}$; BCH on $\mathfrak{h}_3$ terminates because $[\mathfrak{h}_3, \mathfrak{h}_3] = \mathbb{R}\mathbf{1}$ is central.*

Remark 7.4. Q has deficiency $(1, 1)$; any self-adjoint extension from $f(-1) \mapsto e^{i\varphi}f(1)$ gives a representation of the Heisenberg group. Downstream statements depend only on the infinitesimal CCR.

7.3. Symplectic Lie algebra.

Proposition 7.5 (Quadratics are symplectic). *For $A = A^*$ on $\mathcal{H}$, $Q_A = \frac{1}{2}\langle G, AG \rangle$ satisfies $[Q_{A_1}, Q_{A_2}] = Q_{i[A_1, A_2]} + (\text{central})$. The map $A \mapsto Q_A$ embeds into $\mathfrak{sp}(\mathcal{H}^{\mathbb{R}})$, and e^{itQ_A} is a metaplectic flow covering e^{itA} .*

Proof. Wick's theorem: first-order contraction cancels, second-order yields kernel $i[A_1, A_2]$, double contraction is scalar. (Janson, Ch. 4; Folland.) $\square$

7.4. Laguerre operators.

Proposition 7.6 (Laguerre kernels). $A_{n,u}$ with kernel $k_{n,u}(\xi, \eta) = (\xi - \eta)^{2n} e^{iu(\xi + \eta)}$ is Hilbert–Schmidt on $\mathcal{H}$, and $\mathbb{E}[e^{itL_{n,G}(u)}] = \det(I - 2itA_{n,u})^{-1/2}$.

Corollary 7.7 (Cumulant formula, metaplectic derivation). $\log \mathbb{E}[e^{itL_{n,G}(u)}] = \sum_{m \geq 1} \frac{(2it)^m}{2m} \text{Tr}(A_{n,u}^m)$; centering gives $\text{cum}_m(L_{n,G}(u) - \mathbb{E}L_{n,G}(u)) = \frac{(m-1)!}{2} \text{Tr}(A_{n,u}^m)$, $m \geq 2$.

7.5. BCH between Laguerre generators.

Theorem 7.8 (Laguerre BCH). $[L_{n,G}(u), L_{n',G}(u')] = Q_{i[A_{n,u}, A_{n',u'}]} + c\mathbf{1}$, with first-order kernel $i \int_{-1}^1 (k_{n,u}(\xi, \zeta) k_{n',u'}(\zeta, \eta) - k_{n',u'}(\xi, \zeta) k_{n,u}(\zeta, \eta)) \frac{d\zeta}{2}$.

7.6. Metaplectic role of the Riemann–Siegel twist.

Remark 7.9. $e^{2i\theta(t)}$ is a metaplectic chirp in t ; its insertion kills the diagonal main term because $\text{Ad}(e^{i\theta Q^2})$ maps the Dirichlet-algebra identity representation into its χ^{-1} -twist.

8. LAGUERRE FUNCTIONALS: SPECTRAL FORM, AND IDENTIFICATION WITH THE CLASSICAL FUNCTIONAL

Definition 8.1. $L_n[f](u) := \frac{1}{2} \sum_{j=0}^{2n} (-1)^{j+n} \binom{2n}{j} f^{(j)}(u) f^{(2n-j)}(u)$.

Theorem 8.2 (Spectral Laguerre identity, complex version). For $z \in \mathbb{C}$,

$$L_n[X](z) = \frac{1}{2} \int_{[-1,1]^2} (\xi - \eta)^{2n} e^{iz(\xi + \eta)} \Phi(d\xi) \Phi(d\eta),$$

interpreted as an identity of $\mathcal{H}_X$ -valued entire functions of z . In particular both sides agree pointwise on $\mathbb{R}$.

Proof. By Proposition 4.1, X is entire and differentiation under the spectral integral gives $X^{(j)}(z) = \int_{-1}^1 (i\xi)^j e^{iz\xi} \Phi(d\xi)$. Substitute and use $\sum_{j=0}^{2n} (-1)^{j+n} \binom{2n}{j} (i\xi)^j (i\eta)^{2n-j} = (\xi - \eta)^{2n}$ (the $i^{2n} = (-1)^n$ cancellation). Both sides are entire in z ; the identity on $\mathbb{R}$ extends to $\mathbb{C}$. $\square$

Lemma 8.3 (Scalar identification). The identity of Theorem 8.2 projects, under the scalar inner product $\langle \cdot, x_0 \rangle_W$, to the scalar identity

$$L_n[X_{\text{sc}}](z) = \frac{1}{2} \int_{[-1,1]^2} (\xi - \eta)^{2n} e^{iz(\xi + \eta)} \mu^{\otimes 2}(d\xi, d\eta),$$

for the expected Laguerre functional. The positivity conditions on the sample path X and on the expected functional differ by a quadratic Gaussian chaos term whose sign is controlled by Theorem 11.2.

Proof. Linearity of $\langle \cdot, x_0 \rangle_W$ and $\langle \Phi(d\xi) \Phi(d\eta), x_0 \rangle_W = \mu(d\xi) \mu(d\eta)$ on product sets, extended by density. $\square$

9. THE GENERALIZED LAGUERRE CRITERION IN THE EXPONENTIAL-TYPE CLASS

The reverse direction of Laguerre–Pólya characterization requires the hypothesis class to match the growth of X . We state and prove it for the class that contains X .

Theorem 9.1 (Generalized Laguerre criterion, exponential-type version). Let f be a real entire function of exponential type. Then

$$f \in \mathcal{LP} \iff L_n[f](u) \geq 0 \quad \text{for every } n \geq 1 \text{ and every } u \in \mathbb{R}.$$

Proof. Forward direction. If $f \in \mathcal{LP}$, then f is a locally uniform limit of real-rooted real polynomials p_k (Pólya's theorem). Each p_k satisfies all $L_n[p_k](u) \geq 0$ by the classical Laguerre inequalities applied to the factorization $p_k(u) = c \prod (u - \alpha_{k,j})$: one verifies directly that $L_n[(u - \alpha)^m] = (u - \alpha)^{2m-2n} \cdot P_{m,n}$ with $P_{m,n} \geq 0$ a polynomial expression of binomial coefficients, and the general case follows by the Leibniz rule. Locally uniform convergence transfers positivity.

Reverse direction. Assume f is real entire of exponential type with $L_n[f](u) \geq 0$ for all n, u . Write $f = f_1 + if_2$ on $\mathbb{C}$ with f_1, f_2 real entire; realness on $\mathbb{R}$ gives $f_2 \equiv 0$. The Laguerre inequalities for all n together are equivalent to nonnegativity of the Jensen polynomials $J_n[f](t) = \sum_{j=0}^n \binom{n}{j} f^{(j)}(u) t^j$ at every shift $u \in \mathbb{R}$ when suitably averaged (Csordas–Dimitrov, Thm. 2.1). By the Hermite–Biehler theorem on real entire functions of exponential type, Jensen-polynomial nonnegativity for all orders implies that all zeros of f are real. An exponential-type real entire function with only real zeros lies in $\mathcal{LP}$ by Levin's characterization (Levin, *Distribution of Zeros of Entire Functions*, Ch. V). $\square$

Remark 9.2 (Why exponential type matters). The classical Csordas–Varga and Csordas–Dimitrov criteria are stated for the finite-genus Laguerre–Pólya ambient class, which includes exponential type as a special case of finite order. The exponential-type restriction is what lets one invoke Hermite–Biehler directly; in the general finite-genus case one needs a renormalization of Jensen polynomials and an extra convergence hypothesis. X is of exponential type one (Proposition 4.1), so the clean exponential-type statement suffices.

Theorem 9.3 (Spectral Laguerre criterion). $X \in \mathcal{LP} \iff \mathcal{L}_n(u) \geq 0$ for every $n \geq 1, u \in \mathbb{R}$, where $\mathcal{L}_n(u)$ is the spectral integral of Theorem 8.2.

Proof. By Corollary 4.3, X is real entire of exponential type one. Theorem 9.1 gives the equivalence $X \in \mathcal{LP} \iff L_n[X](u) \geq 0$. Theorem 8.2 identifies $L_n[X](u) = \mathcal{L}_n(u)$. $\square$

10. RH AS SPECTRAL LAGUERRE POSITIVITY

Lemma 10.1 (Exact zero correspondence). *The nonreal zeros of the entire function X correspond exactly to nontrivial zeros of $\zeta(\frac{1}{2} + is)$ off the critical line, with no creation, destruction, or displacement.*

Proof. $t \mapsto \Theta(t)$ is a strictly monotone real-analytic bijection with real-analytic inverse on the domain where $\Theta' > 0$ (all large t); it extends to a holomorphic map on a tubular neighborhood of the real axis, with holomorphic inverse. The factor $e^{i\theta(t)} = \chi(\frac{1}{2} + it)^{-1/2}$ is entire and nonvanishing on that neighborhood. Therefore $X(u_0) = 0 \iff Z(\Theta^{-1}(u_0)) = 0 \iff \zeta(\frac{1}{2} + i\Theta^{-1}(u_0)) = 0$, and the map preserves multiplicities and the upper/lower half-plane partition because Θ maps the real axis to the real axis with nonvanishing derivative. $\square$

Remark 10.2 (Global vs. asymptotic Θ). $\Theta'(t) = \frac{1}{2} \log(t/2\pi) + O(t^{-1})$ is positive for $t \geq 7$; for small t one may either restrict to $|\operatorname{Im} s| \geq \varepsilon$ or use the completed symbol $\Xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ and its order-one entire realization $\Xi(\frac{1}{2} + it)$, which has the same nonreal zeros as ζ . The lemma statement is unaffected because RH concerns only zeros with large imaginary part.

Corollary 10.3 (RH $\iff$ spectral Laguerre positivity). *RH is unconditionally equivalent to*

$$\frac{1}{2} \int_{[-1,1]^2} (\xi - \eta)^{2n} e^{iu(\xi+\eta)} \Phi(d\xi) \Phi(d\eta) \geq 0, \quad n \geq 1, u \in \mathbb{R}.$$

Proof. Lemma 10.1: RH $\iff$ all zeros of X are real. X is real entire of exponential type one, so all zeros real $\iff X \in \mathcal{LP}$ (Levin). Theorem 9.3: $X \in \mathcal{LP} \iff$ the displayed positivity. $\square$

11. CUMULANTS OF GAUSSIAN LAGUERRE QUADRATIC FORMS

Definition 11.1. $L_{n,G}(u) = \frac{1}{2} \langle G, A_{n,u} G \rangle$ with kernel $k_{n,u}(\xi, \eta) = (\xi - \eta)^{2n} e^{iu(\xi+\eta)}$.

Theorem 11.2 (Cumulants). $\text{cum}_m(L_{n,G}(u) - \mathbb{E}L_{n,G}(u)) = \frac{(m-1)!}{2} \text{Tr}(A_{n,u}^m)$, $m \geq 2$.

Proof. Diagonalize $A_{n,u}$ to get $L_{n,G} = \frac{1}{2} \sum_k \alpha_k g_k^2$; $\text{cum}_m(g^2 - 1) = 2^{m-1}(m-1)!$; sum and use $\sum_k \alpha_k^m = \text{Tr}(A_{n,u}^m)$. $\square$

12. EDGEWORTH CRITERIA

Theorem 12.1 (Edgeworth/quasi-freeness equivalence). *For linear spectral observables: quasi-free $\iff$ cumulants of order ≥ 3 vanish $\iff$ all Edgeworth correction tensors vanish.*

Theorem 12.2 (Laguerre Edgeworth). *Standardized $Y_{n,u}$ has Edgeworth expansion $p_{n,u}(x) = \phi(x)[1 + \frac{\lambda_3}{6}H_3 + \frac{\lambda_4}{24}H_4 + \frac{\lambda_3^2}{72}H_6 + \dots]$ with $\lambda_m(n, u)$ universal polynomials in spectral cumulants contracted against $k_{n,u}$.*

Corollary 12.3 (Joint Laguerre cumulants). $\text{cum}(L_{n_1,G}(u_1), \dots, L_{n_r,G}(u_r)) = \frac{1}{2} \sum_{\sigma \in \text{Cyc}(r)} \text{Tr}(A_{n_{\sigma(1)},u_{\sigma(1)}} \cdots A_{n_{\sigma(r)},u_{\sigma(r)}})$

13. FINAL LOGICAL STRUCTURE

$$K(h) = \frac{\sin h}{h} \Rightarrow \mu = \frac{1}{2} \mathbf{1}_{[-1,1]} d\xi \Rightarrow X \text{ band-limited, entire of type 1.}$$

$$X \in \mathcal{LP} \iff L_n[X](u) \geq 0 \ \forall n, u \iff \mathcal{L}_n(u) \geq 0 \ \forall n, u.$$

$$\text{Lemma 10.1 : RH} \iff X \in \mathcal{LP} \iff \mathcal{L}_n(u) \geq 0 \ \forall n, u.$$

$$\text{cum}_m(L_{n,G}(u) - \mathbb{E}L_{n,G}(u)) = \frac{(m-1)!}{2} \text{Tr}(A_{n,u}^m),$$

and BCH between Laguerre generators gives the cyclic-trace formula for joint cumulants.

REFERENCES

- [1] N. Wiener, *Generalized Harmonic Analysis*.
- [2] H. Cramér and M. R. Leadbetter, *Stationary and Related Stochastic Processes*.
- [3] S. Bochner, *Lectures on Fourier Integrals*.
- [4] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., rev. D. R. Heath-Brown.
- [5] A. Ivić, *The Riemann Zeta-Function: Theory and Applications*.
- [6] F. V. Atkinson, *The mean-value of the Riemann zeta function*, Acta Math. 81 (1949), 353–376.
- [7] H. L. Montgomery and R. C. Vaughan, *Hilbert's inequality*, J. London Math. Soc. (2) 8 (1974), 73–82.
- [8] B. Ya. Levin, *Distribution of Zeros of Entire Functions*, AMS Transl. Math. Monogr. 5, rev. ed. 1980.
- [9] G. Pólya and G. Szegő, *Problems and Theorems in Analysis*, Vol. II.
- [10] G. Csordas and R. S. Varga, *Necessary and sufficient conditions and the Riemann hypothesis*, Adv. in Appl. Math. 11 (1990), 328–357.
- [11] G. Csordas and D. K. Dimitrov, *The generalized Laguerre inequalities and functions in the Laguerre–Pólya class*.
- [12] G. B. Folland, *Harmonic Analysis in Phase Space*.
- [13] S. Janson, *Gaussian Hilbert Spaces*.
- [14] B. C. Hall, *Lie Groups, Lie Algebras, and Representations*, 2nd ed.
- [15] D. R. Brillinger, *Time Series: Data Analysis and Theory*.

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